

Zehao Dang

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Education

Virginia Tech B.S. in Computer Engineering, with Control, Robotics and Autonomy Concentration (GPA: 3.47 / 4.00) • Relevant Coursework: Introduction to Control Systems, Principles of Robotics, Machine Learning	Aug 2025 - May 2027 Blacksburg, VA
Sichuan University B.E. in Electrical and Computer Engineering (GPA: 3.73 / 4.00) • Relevant Coursework: Data Structures and Algorithms (C++), Signals and Systems, Embedded Systems	Aug 2023 - May 2025 Chengdu, P.R. China

Employment

Grado Department of Industrial and Systems Engineering, Virginia Tech <i>Research Assistant (Paid, Part-Time)</i> • Leading the development of a vision-based adaptive 3D-printing system for dynamically deforming surfaces, integrating mechanical design, prototyping, Raspberry Pi control, computer vision and predictive feedback control; targeting submission to NAMRC 2027. • Led a team of 4 undergraduate researchers to develop an end-to-end robotic additive manufacturing pipeline converting Cura G-code into UR5e trajectories using inverse kinematics, redundancy-aware path optimization, and trajectory retiming, with simulation validation in NVIDIA Isaac Sim.	Apr 2026 – Present
Hi-Tech Optoelectronics Co., Ltd. <i>Product Quality Intern (Full-Time)</i> • Inspected semiconductor laser chip modules under optical microscopes to identify manufacturing defects and verify product quality. Supported production-line operations by assembling and testing power-supply units for diode-pumped laser systems.	Jul 2024 – Aug 2024 Beijing, P.R. China

Experience

AutoNav Team, VT CRO (Advisor: Dr. Matt Nowinski) <i>Robotic Software Engineer</i> • Building a ROS2/Gazebo simulation environment for an autonomous ground vehicle competing in the 34th Intelligent Ground Vehicle Competition (IGVC) in June 2027, including sensor simulation and navigation-stack validation. • Developing reproducible test scenarios to evaluate localization, path planning, and obstacle-avoidance algorithms.	Aug 2026 – Present
Math Department, Sichuan University - Pittsburgh Institute (PI: Dr. Zheng Yang) <i>Undergraduate Researcher</i> • Assisted in evaluating machine-learning models for AD, MCI, and NC classification by comparing test accuracy across classifiers. • Conducted literature review on machine-learning-based Alzheimer's disease classification and contributed to manuscript writing.	May 2025 – Aug 2025
ECE Department, Sichuan University - Pittsburgh Institute (PI: Dr. Jeungphill Hanne) <i>Undergraduate Researcher</i> • Conducted a literature review of analytical methods for infrared thermography-based bone-fracture diagnosis. Coordinated biweekly research-paper presentations for a team of six undergraduate researchers.	Mar 2024 – May 2025

Publications

- Siyuan Liu, Cheng Sun, Hangcheng Wu, **Zehao Dang**, Yucong Wei, Zheng Yang, Kang Li, Peng Chen, Yang Lv, and Wenbo Zhang, "A Topological Classification for Alzheimer's Disease via Hemoglobin Concentration fNIRS Data," *International Conference on Industrial Automation, Robotics and Control Engineering (IARCE)*, 2025. [doi:10.1109/IARCE68366.2025.11485743](https://doi.org/10.1109/IARCE68366.2025.11485743)

Technical Skills

Robotics	NVIDIA Isaac Sim, ROS2, Gazebo
Software	Git, Python, C++, Linux, Qt, CMake, Docker, MATLAB
Hardware	CAD, 3D Printing, Arduino, Raspberry Pi, LTspice
Technical Documentation	LaTeX, Markdown, Microsoft Office

Honors and Awards

Dean's List, College of Engineering, Virginia Tech	Spring 2026
2nd Class Scholarship, Sichuan University	2024, 2025

Standardized Test Scores

GRE	Quantitative 170, Verbal 154, Analytical Writing 3.5
IELTS	Overall 8.0 , Listening 9.0, Reading 8.5, Writing 8.0, Speaking 7.0